Questions of the day

- We have seen a couple of sorting algorithms so far
  - Are there faster sorting algorithms?
  - What is the fastest *possible* sorting algorithm?

Sorting and selection algorithms

William Hendrix

Lecture 5

Today

- Review
  - Data structures
  - Sorting algorithms
- BubbleSort
- MergeSort
- QuickSort
- Average-case analysis
- Comparison-based sorting lower bound
- Non-comparison-based sorting algorithms
  - CountingSort
  - BucketSort
  - RadixSort
- QuickSelect
- RSelect

Review: data structures

- *For all:* know operations, implementation, and complexity
- **Stack:** access elements in LIFO order
- **Queue:** access elements in FIFO order
  - **Deque:** can simulate stack or queue
  - Array preferred, linked list okay
- **Priority queue:** access elements in "best first" order
  - Heap or Fibonacci heap
- **List:** store elements by index
  - Array or linked list
- **Set:** searchable collection of objects
- **Map:** searchable collection of associations (*key-value pairs*)
  - Balanced BST, hash table, or array
- **Union-Find:** represents partition/group memberships
  - Union by rank and path compression

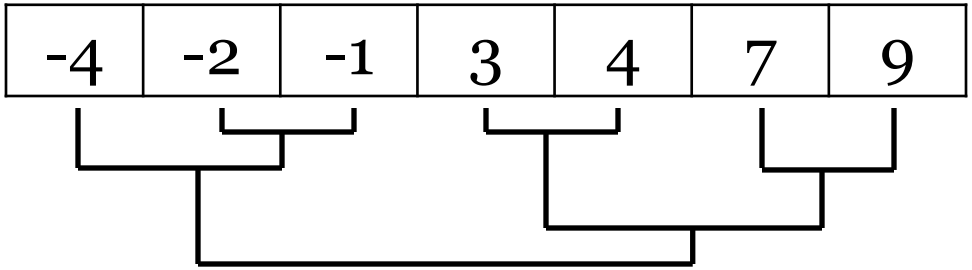
Union-Find application

- Major application: algorithm later in semester…
- <u>Single linkage hierarchical clustering</u>
  - All points start as separate, individual clusters (groups)
  - Repeatedly join "closest" clusters together
  - *Single linkage:* distance between clusters = min dist b/w points
- **Input:**
  - *data:* array of *n* points
  - *c:* number of clusters
- **Output:** *c* clusters
- **Pseudocode:**

```
1. Initialize union-find
2. Calculate all distances between points
3. Repeat:
4.     If points with next smallest distance are not in same group:
5.         Union them
6. Until there are c groups in union-find
```

Hierarchical clustering example

- **Pseudocode**
  - Union next closest points if not unioned
  - Repeat until desired # of clusters
- **Example:** *data* are integers, distance is abs. value, $c = 1$



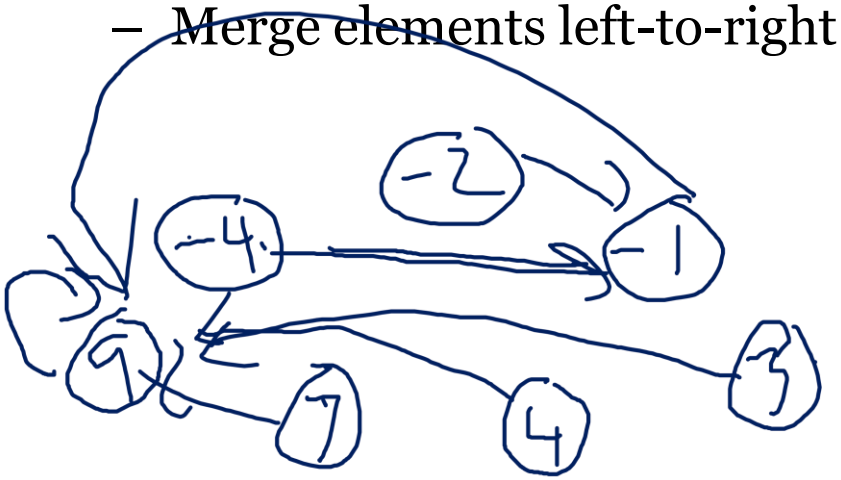
dist = 1
dist = 2
dist = 3
dist = 4

Union-Find exercise

- Perform single-linkage hierarchical clustering on the array below until everything is in one cluster
  - Distance is abs. value
- Draw the Union-Find data structure after each step

-4	-2	-1	3	4	7	9

- Rules
  - If given a choice, merge the cluster with the leftmost element
  - Merge elements left-to-right



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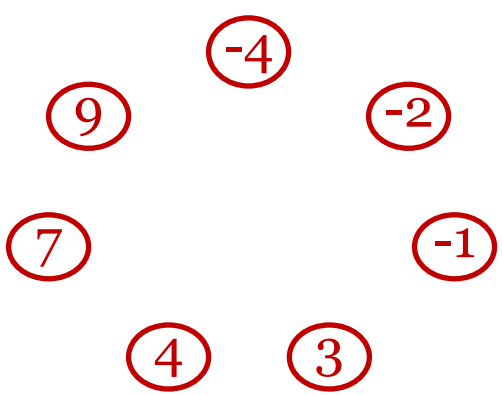
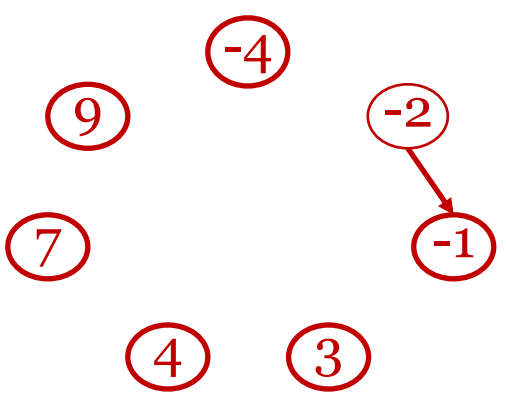
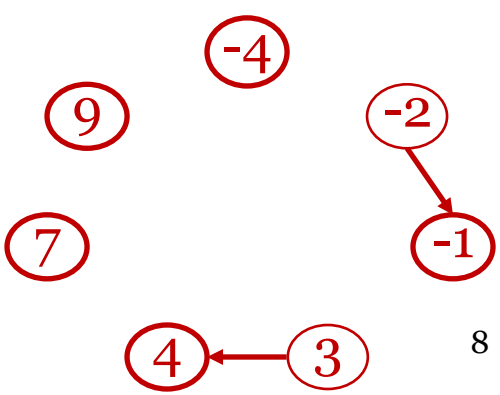
Initialize



Union(2, 3)
or: Union(-2, -1)



Union(4, 5)
or: Union(3, 4)



Union-Find exercise

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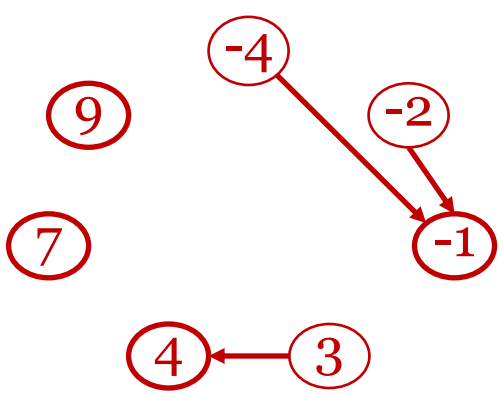
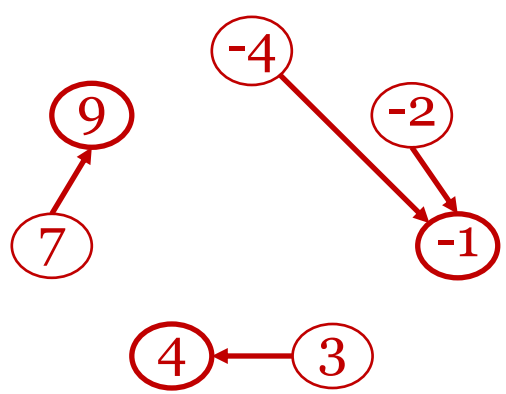
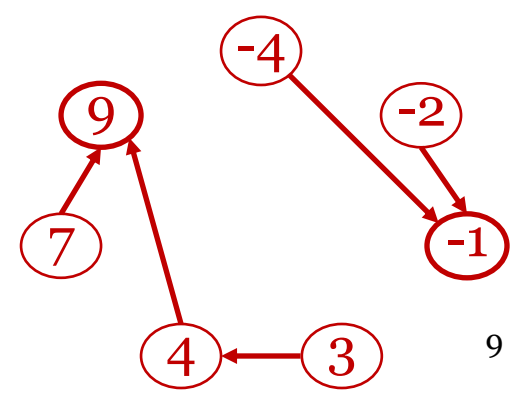
Union(1, 2)
or: Union(-4, -2)



Union(6, 7)
or: Union(7, 9)



Union(5, 6)
or: Union(4, 7)



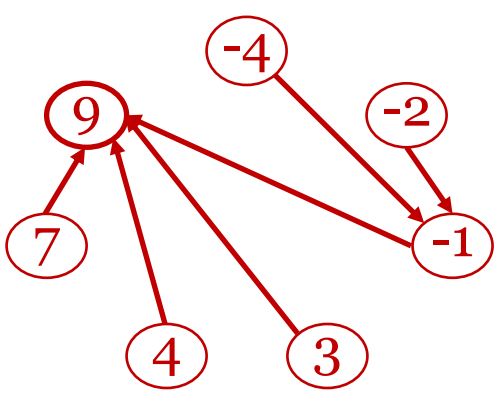
Union-Find exercise

- Perform single-linkage hierarchical clustering on the array below until everything is in one cluster
  - Distance is abs. value
- Draw the Union-Find data structure after each step

-4	-2	-1	3	4	7	9

- Rules
  - If given a choice, merge the cluster with the leftmost element
  - Merge elements left-to-right

Union(3, 4)
or: Union(-1, 3)



Array representation (final):

3	3	7	7	7	7	7

Note that array values represent the *position* (index) of the element, not its value.
For example, element #1 (-4) points to element #3 (-1), which points to element #7 (9).

Union-Find complexity

- **Find(x)**
  - Recursively point to answer
  - $\Theta(\alpha(n))$, amortized
  - Generally less than 4 for conceivable n
- **Union(a, b)**
  - Call Find on both sides first
  - Always point to larger tree
  - $\Theta(\alpha(n))$

```
Algorithm: Find(x)
if unionfind[x] ≠ x then
    id = Find(unionfind[x])
    unionfind[x] = id
end
return unionfind[x]
```

```
Algorithm: Union(a, b)
ra = Find(a)
rb = Find(b)
if size[ra] > size[rb] then
    Swap ra and rb
end
unionfind[ra] = rb
size[rb] = size[ra] + size[rb]
```

- The Ackermann function
  - Incredibly fast-growing function
  - First few values:
    - $3, 7, 61, 2^{2^{2^{65536}}} - 3, \ldots$

Other worthwhile data structures

- **Prefix tree**
  - A.k.a., trie
  - Stores a collection of strings
  - Can efficiently search whether a string begins with query
    - Spellchecking or auto-complete
  - *Notable variants:* compressed prefix tree, suffix tree

- ***k*-d tree**
  - "k-dimensional tree"
  - Like BST for spatial data (2D, 3D, etc.)
  - Supports efficient nearest neighbor searching
  - Constructed to be perfectly balanced
    - Does not self-balance
  - *Alternatives:* quad-trees (2D data) or oct-trees (3D)

Review: sorting algorithms

- **SelectionSort**
  - Iteratively swap min into next spot
  - Best/worse case complexity: $\Theta(n^2)$

- **InsertionSort**
  - Iteratively insert next value into sorted array
  - Best/worse case complexity: $\Omega(n)/O(n^2)$

- **HeapSort**
  - Similar to SelectionSort but with heaps
  - Heapify array (max heap)
  - DeleteMax() $n - 1$ times
  - Best/worse case complexity: $\Theta(n \lg n)$

BubbleSort

- Another iterative sorting algorithm
- Scan array:
  - Swap adjacent elements if out of order
- Repeat until no swaps occur

```
Input: data: array of integers to sort
Input: n: the size of data
Output: permutation of data such that
        data[1] ≤ ... ≤ data[n]
Algorithm: BubbleSort
repeat
    for i = 1 to n − 1 do
        if data[i] > data[i + 1] then
            Swap data[i] and data[i + 1]
        end
    end
until the for loop makes no swaps
return data
```

BubbleSort analysis

$\Omega(n)$

$\Omega(1)$ iter

Input: *data*: array of integers to sort
Input: n: the size of *data*
Output: permutation of *data* such that
$data[1] \leq \ldots \leq data[n]$

```
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$\Theta(1)$ iter

$\Theta(1)$

$\Theta(n)$ iters $\Rightarrow \Theta(n)$ /iter

BubbleSort analysis

Input: $data$: array of integers to sort
Input: n: the size of $data$
Output: permutation of $data$ such that
$data[1] \leq \ldots \leq data[n]$

Algorithm: BubbleSort

repeat
  for $i = 1$ to $n - 1$ **do**
    if $data[i] > data[i+1]$ **then**
      Swap $data[i]$ and $data[i+1]$
    end
  end
until the for loop makes no swaps
return $data$

$\Theta(n)$ iters

$\Theta(1)$

BubbleSort analysis

```
Input: data: array of integers to sort
Input: n: the size of data
Output: permutation of data such that
        data[1] ≤ ... ≤ data[n]
Algorithm: BubbleSort
repeat                                   O(n)
    for i = 1 to n − 1 do                Θ(n)
        if data[i] > data[i + 1] then
            Swap data[i] and data[i + 1]
        end
    end
until the for loop makes no swaps
return data
```

After iteration k of the outer loop, the last k values will be the largest k values in the array, in sorted order

BubbleSort analysis

```
Input: data: array of integers to sort
Input: n: the size of data
Output: permutation of data such that
        data[1] ≤ ... ≤ data[n]
Algorithm: BubbleSort
repeat                                  O(n²)
    for i = 1 to n − 1 do
        if data[i] > data[i + 1] then
            Swap data[i] and data[i + 1]
        end
    end
until the for loop makes no swaps
return data                             Θ(1)
```

Total: $O(n^2)$

After iteration k of the outer loop, the last k values will be the largest k values in the array, in sorted order

MergeSort

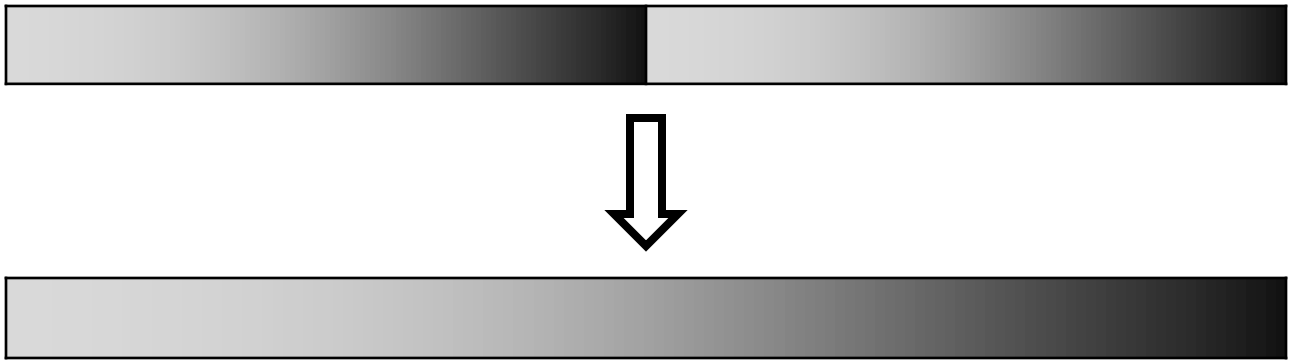
- Fast algorithm for sorting
  - InsertionSort and SelectionSort are $O(n^2)$

<u>Description</u>

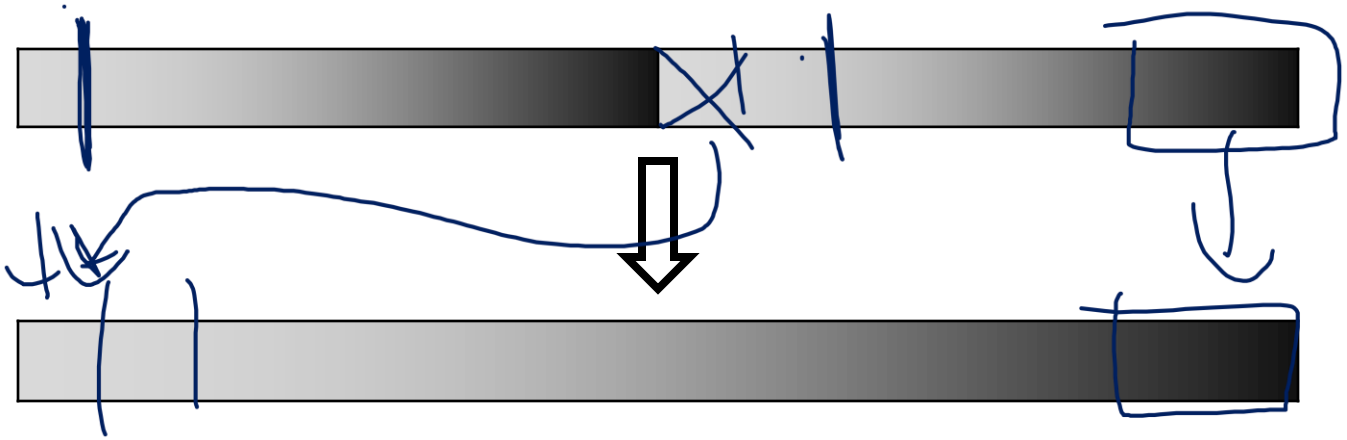
- Split array into two halves

L	**R**

  - Sort half-arrays recursively
    - Base case: one element
  - Combine two sorted half-arrays into one sorted array



Merging



- Fill in result left-to-right
- Min is min(left) or min(right)
- 2nd value is next value of selected half or min(unselected half)
- Continue until both arrays have emptied into result
  - After one array is empty, just add the other
- Time to combine: $\Theta(n)$
  - Better than $O(n^2)$

MergeSort pseudocode

Input: *data*: the data to sort (must be comparable)
Input: n: the number of elements in data
Output: a permutation of *data* such that $data[1] \leq \ldots \leq data[n]$
Algorithm: MergeSort

```
if n ≤ 1 then
    return data;
end
mid = floor((n + 1)/2);
left = MergeSort(data[1..mid], mid);
right = MergeSort(data[mid + 1..n], n − mid);
temp = Array(n);
ℓ = r = 1;
while ℓ ≤ mid and r ≤ n − mid do
    if left[ℓ] < right[r] then
        temp[ℓ + r − 1] = left[ℓ];
        ℓ = ℓ + 1;
    else
        temp[ℓ + r − 1] = right[r];
        r = r + 1;
    end
end
temp[ℓ + r − 1..mid + r − 1] = left[ℓ..mid];
temp[mid + r..n] = right[r..n − mid];
return temp;
```

$$T(n) = 2T(n/2) + \Theta(n)$$

$$T(n) = \Theta(n \lg n)$$

QuickSort

- Another $n \log n$ algorithm for sorting
- Splits dataset according to *value*, not *position*
  - "Small half": less than some value
  - "Large half": larger than some value
  - *Pivot*: the value used to split the dataset
- Pseudocode
  1. Select pivot
     - Naïve strategy: pick first element
  2. Start at left and right ends of array
     - Swap values larger than pivot to the RHS of array
     - Swap values smaller than pivot to LHS of array
     - Equal values can go on either side
  3. Insert pivot where two "halves" meet
  4. Recursively sort each "half"
     - Base case: arrays with 0 or 1 elements are sorted

QuickSort example

- Apply QuickSort to the array below:
  1. Select pivot
     - Naïve strategy: pick first element

75	5	55	88	78	42	18	93

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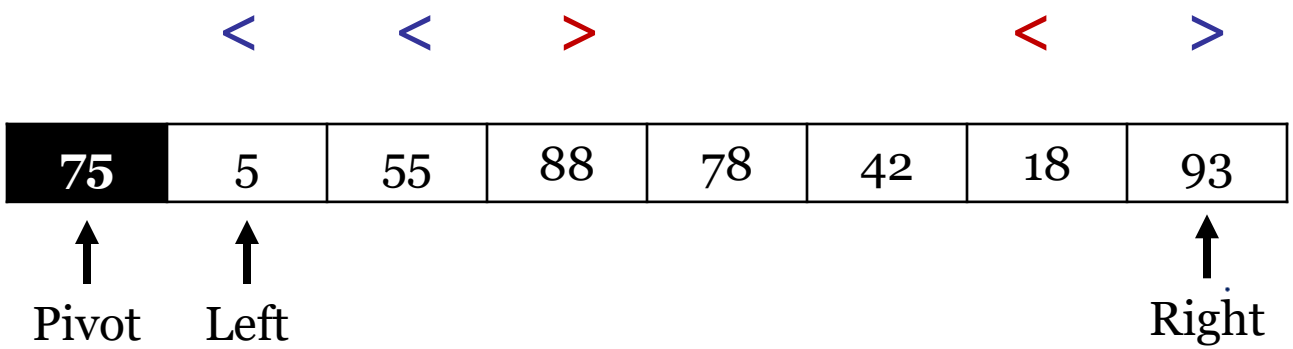

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	<	<	<	>	<	>	>
75	5	55	18	78	42	88	93
Pivot			Left			Right	

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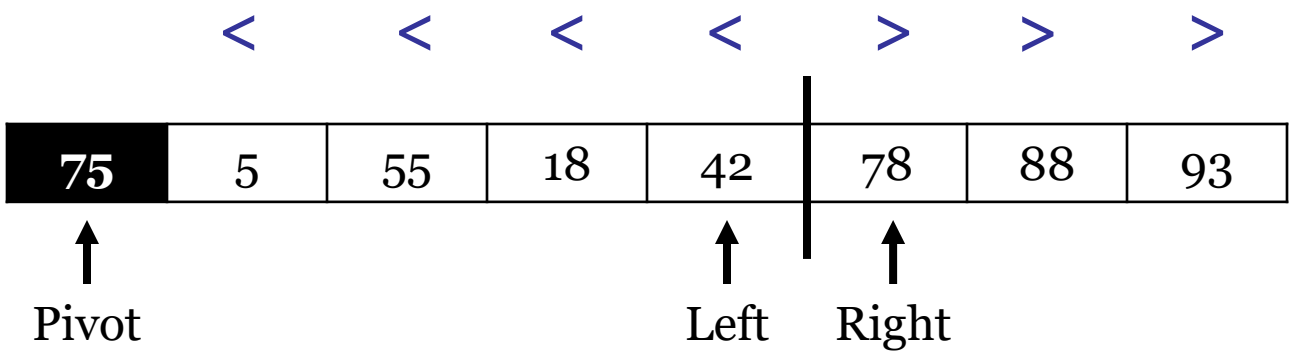

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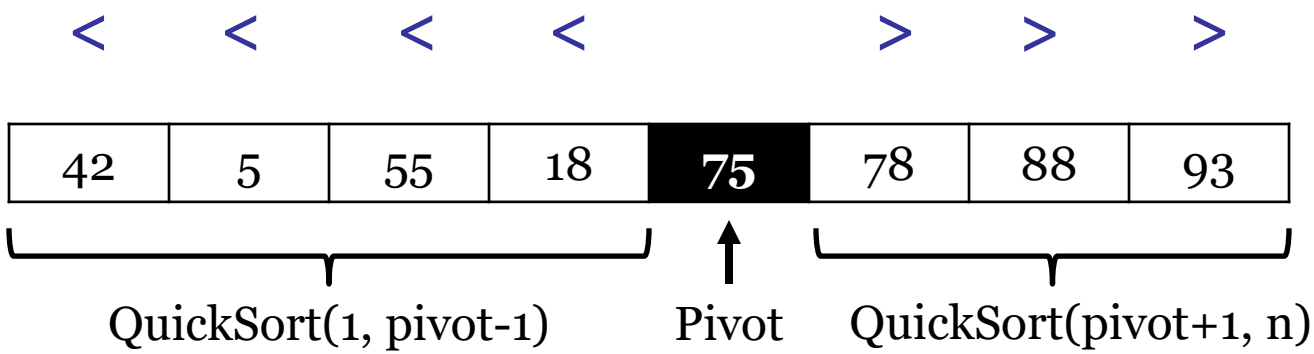
  4. Recurse on both sides



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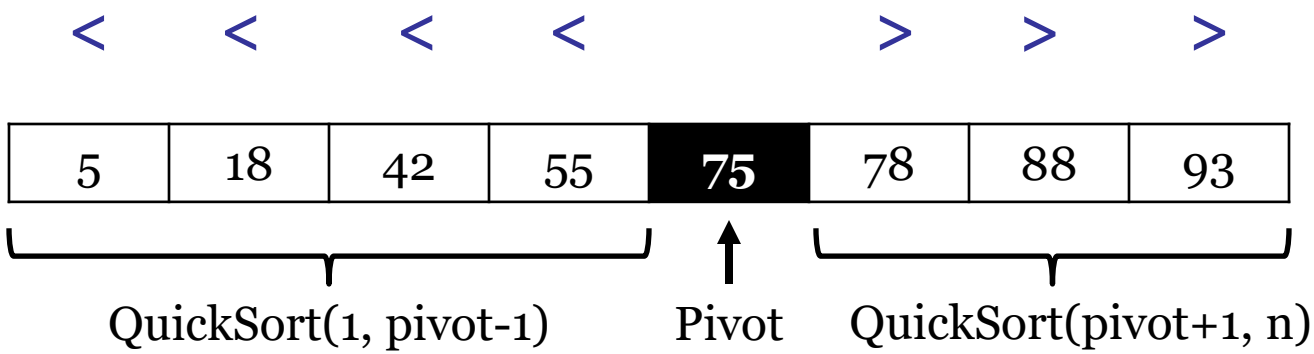
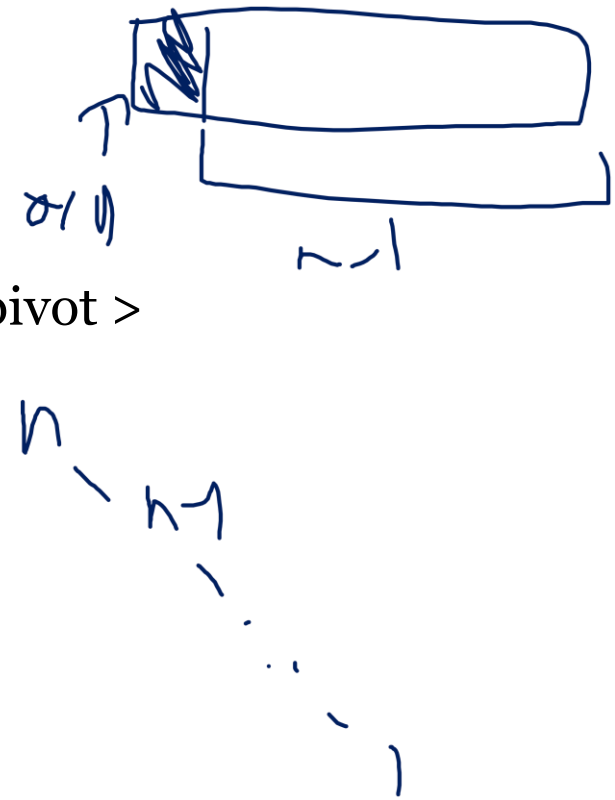


QuickSort complexity

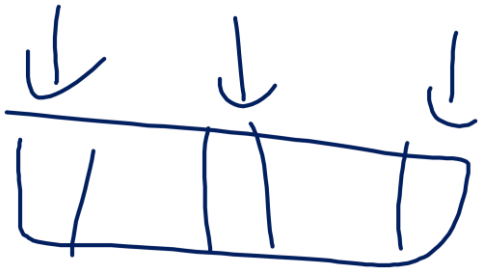
1. Choose pivot
   - Different strategies
2. Swap elements
   - Everything left of pivot is <, right of pivot >
3. Recurse on both sides



- Step 1: $\Theta(1)$
- Step 2: $\Theta(n)$
- Recursion
  - Depends on pivot!
  - $T(n) = T(L) + T(R) + \Theta(n)$
- Worst case: $T(n) = T(n-1) + \Theta(n)$
  - $O(n^2)$
- Best case: $T(n) = 2T(n/2) + \Theta(n)$
  - $\Omega(n \lg n)$

Analysis of QuickSort

- Once the two halves are sorted, entire array is sorted
  - Can use tail recursion
- Pivot might not divide dataset exactly in half
- Pivot value impacts runtime
- Pivot selection strategies
  - First/last
    - Simple
    - Bad on sorted or constant data
  - Median-of-three
    - Choose median of first, last, and middle
    - Partitions sorted data well
  - Random
    - Always average case, unless constant values
  - Median
    - Overhead of calculating is too high: $\Theta(n)$
- Constant data can be fixed by splitting array into <, =, and >
  - Code is more complex



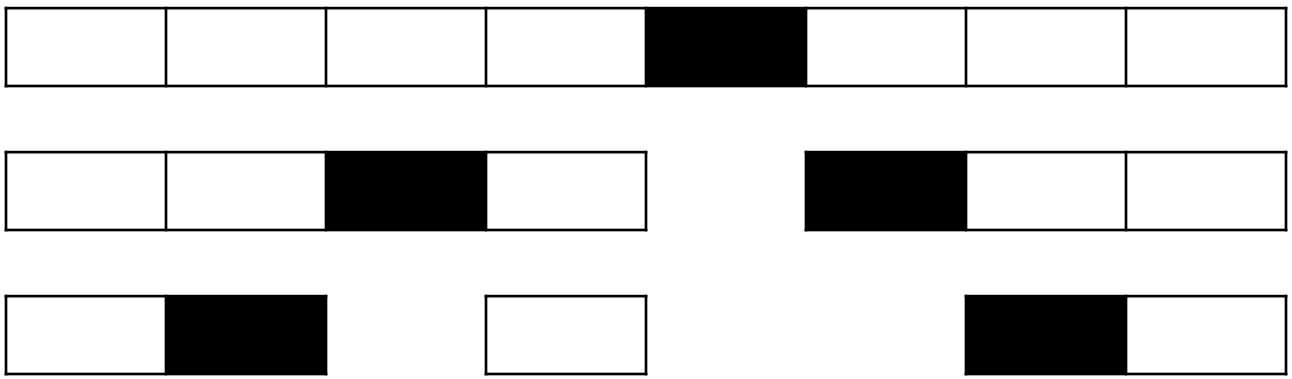
Average/expected case complexity

- Best-case complexity
  - Least amount of time to compute
  - E.g., InsertionSort is $\Theta(n)$ when input is sorted
- Worst-case complexity
  - Greatest amount of time to compute
  - E.g., InsertionSort is $\Theta(n^2)$ when input is reverse-sorted

- Average-case complexity
  - Average complexity across all possible inputs
  - Always falls between best and worst case
  - E.g., inner loop of InsertionSort needs to shift $n/2$ elements on average: $\Theta(n^2/2) = \Theta(n^2)$
- Expected-case complexity
  - Similar to average-case, but with assumptions about inputs

Average-case complexity of QuickSort

- Steps 1 & 2: $\Theta(n)$
- Recursion tree



Complexity

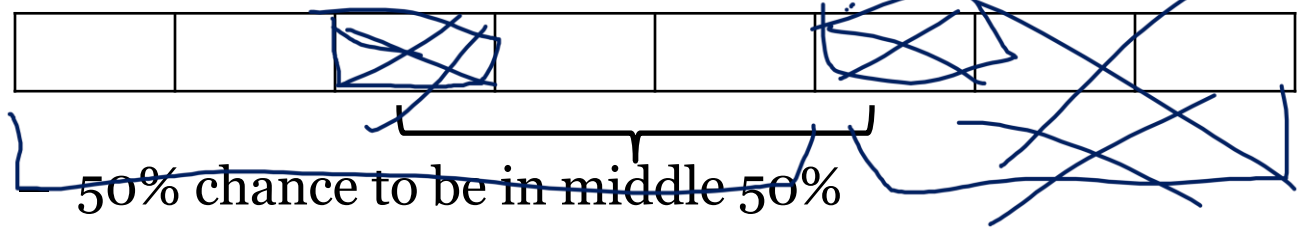
$\Theta(n)$

$O(n)$

$O(n)$

$\vdots$

- Total complexity: $O(nh)$
- Where is the pivot?



  - 50% chance to be in middle 50%
  - Reduces "big" side to ¾ $h = \log_{4/3}(n)$
  - If other pivots do nothing: $h = 2\log_{4/3}(n) \Rightarrow O(n \lg n)$

Comparison-based sorting algorithms

- Sorting algorithms that operate by comparing pairs of elements
  - Everything we've discussed so far
  - All $O(n^2)$ or $O(n \lg n)$

- Is there a faster sorting algorithm?
  - No!
  - At least not comparison-based

- **Proof idea**
  - Based on *execution paths* of algorithm
    - Instructions executed for a given input
    - If statements, loop conditions, etc.
  - Execution path defines all comparisons and swaps
  - Execution path must be different for different inputs

CBS algorithm lower-bound

- Array of size n has $n!$ total permutations
  - Min # of execution paths to be correct
- Each comparison: 2 outcomes
  - true or false
- # of execution paths after k comparisons:

Comparisons	Picture	Count

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Comparisons	Picture	Count
0		1

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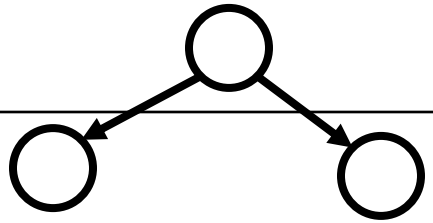
Comparisons	Picture	Count
0		1
1		2

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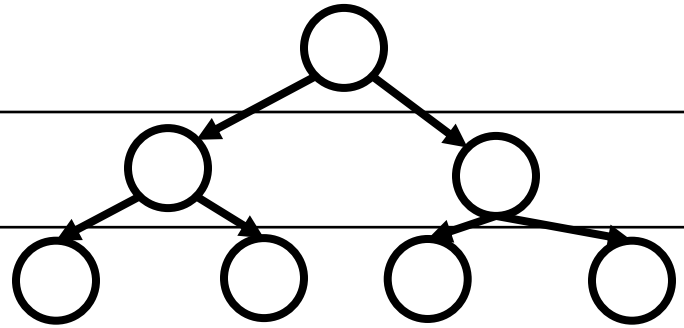
Comparisons	Picture	Count
0		1
1		2
2		4

CBS algorithm lower-bound

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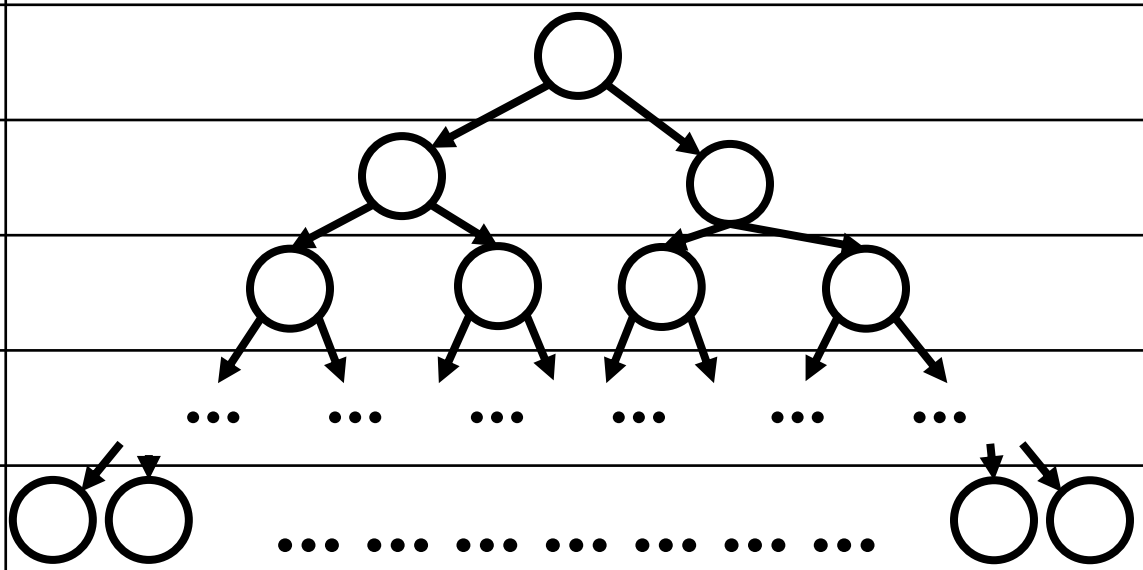
- Each comparison: 2 outcomes
  - true or false
- # of execution paths after k comparisons:

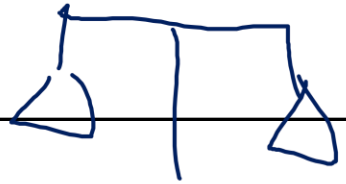
$2^k \geq n!$

$k =$

Comparisons	Picture	Count
0		1
1		2
2		4
…		…
k		2^k

- # comparisons to distinguish $n!$ outcomes: $\lg(n!) = \Theta(n \lg n)$

Lower-bound exercise



- The *counterfeit coin* problem:
  - **Input:** a stack of n coins, all identical except for one counterfeit, which weighs less
  - **Output:** index of counterfeit coin
  - *Primary operation:* weigh any two sets of coins on a scale
    - Returns <, >, or =

1. What is the number of possible outputs for an input of size n?

   $\boxed{n}$

2. How many execution paths do k weighings yield?

   $\boxed{3^k}$

3. Find a lower bound on the weighings required for this problem.

   $k \geq \log_3(n)$

Lower-bound exercise

- The *counterfeit coin* problem:
  - **Input:** a stack of n coins, all identical except for one counterfeit, which weighs less
  - **Output:** index of counterfeit coin
  - *Primary operation:* weigh any two sets of coins on a scale
    - Returns <, >, or =

1. What is the number of possible outputs for an input of size n?
   - n (any of the n coins could be the counterfeit)
2. How many execution paths do k weighings yield?
   - 3^k (each has 3 possibilities)
3. Find a lower bound on the weighings required for this problem.
   - $3^k \geq n$
   - $k \geq \log_3(n)$

- **Food for thought:** can you find the optimum algorithm that makes $\log_3(n)$ comparisons?

Non-comparison-based sorting

- **Main idea**
  - Locate the correct position without comparing to other values in array
  - Compare values to constants rather than each other
  - Use additional memory to avoid direct comparisons

- Three main algorithms
  - CountingSort
  - BucketSort
  - RadixSort

- Each has different advantages/disadvantages
  - None are unconditionally better than comparison-based

CountingSort

3 1 4 2 5

3 4 8 10 15

- **Main idea**
  - For each element:
    - Count number of smaller elements in array
    - Drop into correct position in sorted array
  - Data must be nonnegative integers
    - Negative values more difficult

- **Pseudocode**
  - Find max(*data*)
  - Create array *count* of length max(*data*)
  - Increment count so that *count*[*i*] is # of *i*'s in *data*
  - Prepend 0 to count array
  - Perform a cumulative sum
    - *count[i]* is number of values < *i* (where *i* goes in sorted order)
  - Place each *data[i]* into correct position according to *count*

CountingSort example

```
Input: data: array of n values
Input: n: size of data
Output: a permutation of data such that
        data[0] ≤ ... ≤ data[n − 1]
Algorithm: CountingSort
max = max(data)
count = Array(max)
Initialize count to 0
for i = 0 to n − 1 do
    Increment count[data[i]]
end
Perform a cumulative sum on count
sorted = Array(n)
for i = n − 1 down to 0 do
    Decrement count[data[i]]
    sorted[count[data[i]]] = data[i]
end
return sorted
```

2	1	3	2	2	1

* Pseudocode uses 0 base and assumes *data* > 0

CountingSort example

Input: *data*: array of n values
Input: n: size of *data*
Output: a permutation of *data* such that
$data[0] \leq \ldots \leq data[n-1]$

```
Algorithm: CountingSort
max = max(data)
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Initialize count to 0
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sorted = Array(n)
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    Decrement count[data[i]]
    sorted[count[data[i]]] = data[i]
end
return sorted
```

2	1	3	2	2	1

count:

0	0	0	0
0	1	2	3

* Pseudocode uses 0 base and assumes *data* > 0

CountingSort example

Input: *data*: array of n values
Input: n: size of *data*
Output: a permutation of *data* such that
$data[0] \leq \ldots \leq data[n-1]$

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Algorithm: CountingSort
max = max(data)
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    Decrement count[data[i]]
    sorted[count[data[i]]] = data[i]
end
return sorted
```

2	1	3	2	2	1

count:

0	2	3	1
0	1	2	3

CountingSort example

Input: *data*: array of n values
Input: n: size of *data*
Output: a permutation of *data* such that
$data[0] \leq \ldots \leq data[n-1]$

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end
Perform a cumulative sum on count
sorted = Array(n)
for i = n − 1 down to 0 do
    Decrement count[data[i]]
    sorted[count[data[i]]] = data[i]
end
return sorted
```

2	1	3	2	2	1

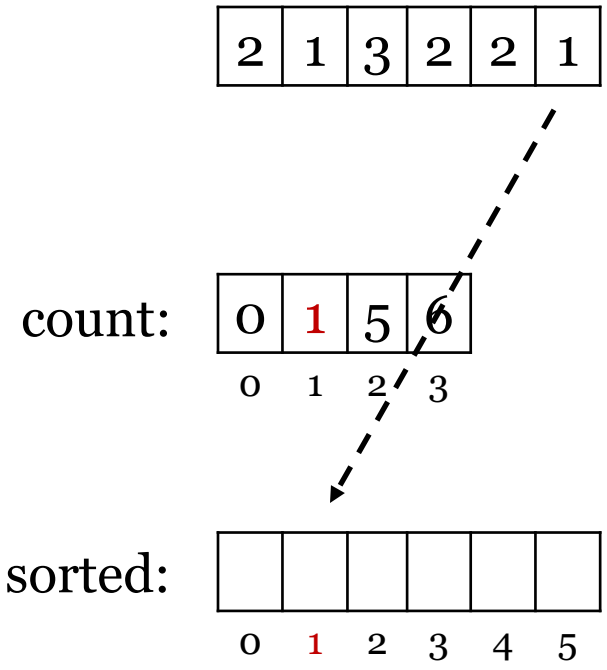
count:

0	2	5	6
0	1	2	3

CountingSort example

Input: *data*: array of n values
Input: n: size of *data*
Output: a permutation of *data* such that
$data[0] \leq \ldots \leq data[n-1]$

Algorithm: CountingSort
$max = \max(data)$
$count = \text{Array}(max)$
Initialize *count* to 0
for $i = 0$ to $n-1$ **do**
  Increment $count[data[i]]$
end
Perform a cumulative sum on *count*
$sorted = \text{Array}(n)$
for $i = n-1$ down to 0 **do**
  Decrement $count[data[i]]$
  $sorted[count[data[i]]] = data[i]$
end
return *sorted*



CountingSort example

Input: *data*: array of n values
Input: n: size of *data*
Output: a permutation of *data* such that
$data[0] \leq \ldots \leq data[n-1]$

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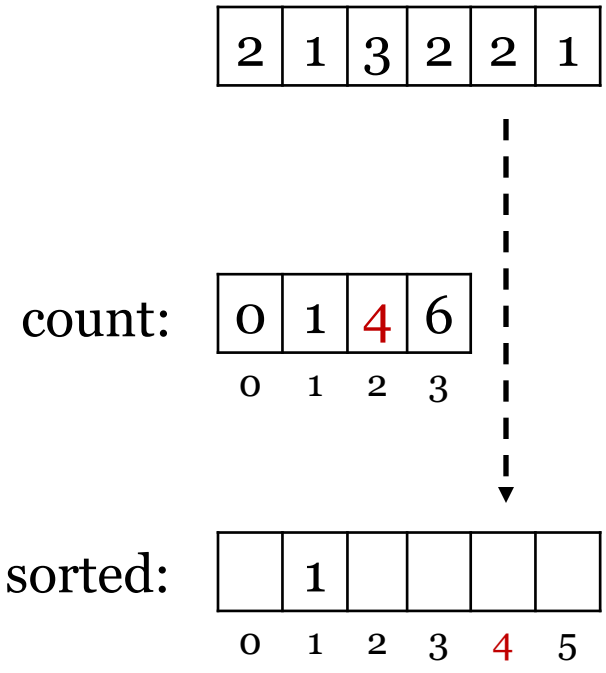

CountingSort example

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Output: a permutation of *data* such that
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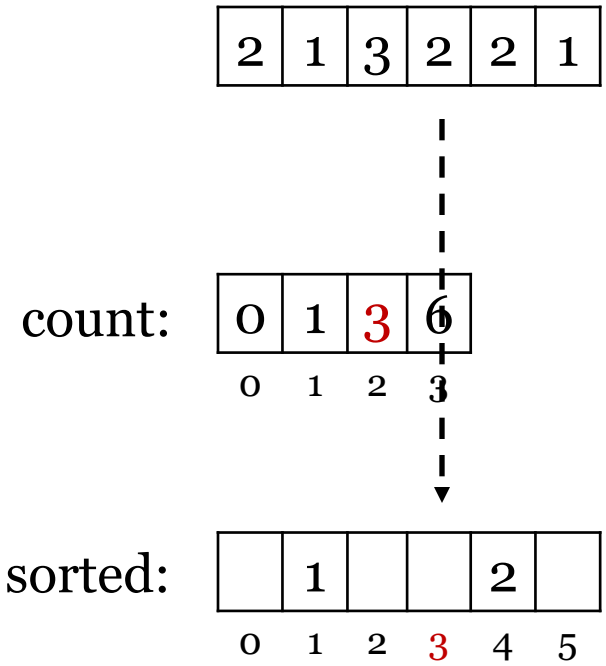

CountingSort example

Input: *data*: array of n values
Input: n: size of *data*
Output: a permutation of *data* such that
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Algorithm: CountingSort
$max = \max(data)$
$count = \text{Array}(max)$
Initialize *count* to 0
for $i = 0$ to $n-1$ **do**
  Increment $count[data[i]]$
end
Perform a cumulative sum on *count*
$sorted = \text{Array}(n)$
for $i = n-1$ down to 0 **do**
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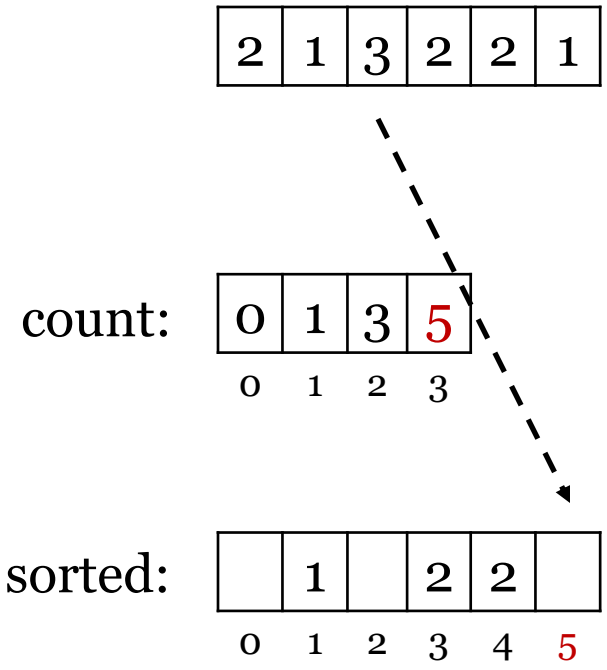
  $sorted[count[data[i]]] = data[i]$
end
return *sorted*



CountingSort example

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Output: a permutation of *data* such that
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Algorithm: CountingSort
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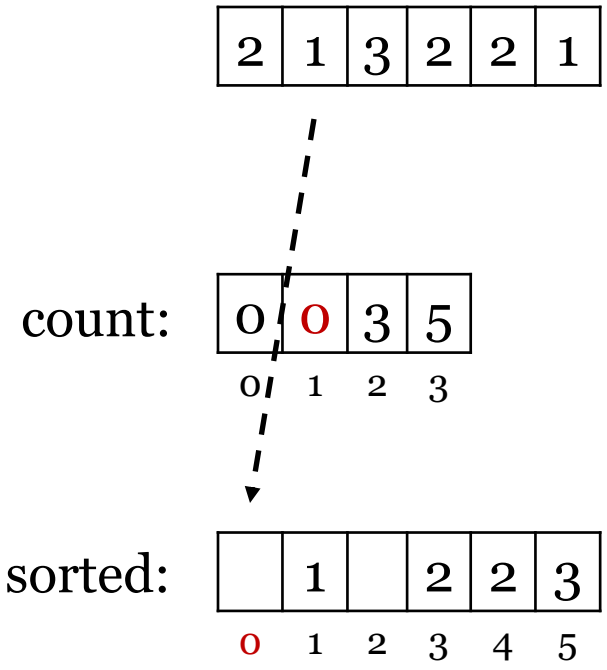
    sorted[count[data[i]]] = data[i]
end
return sorted
```



CountingSort example

Input: *data*: array of n values
Input: n: size of *data*
Output: a permutation of *data* such that
$data[0] \leq \ldots \leq data[n-1]$

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Algorithm: CountingSort
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end
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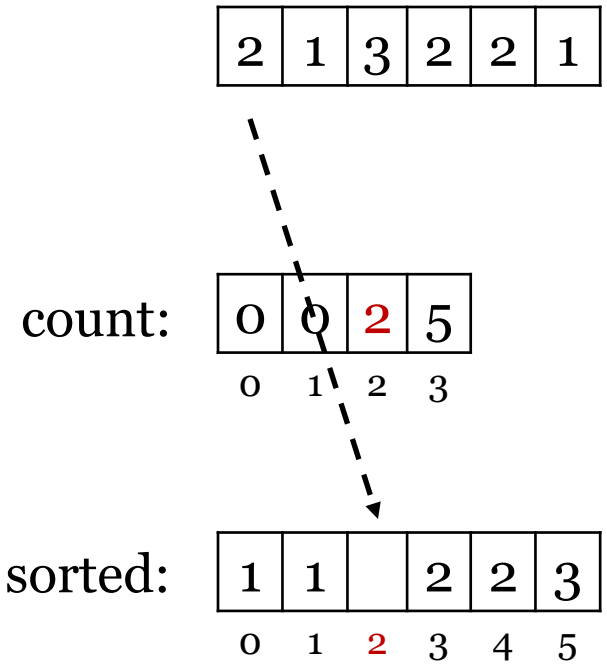
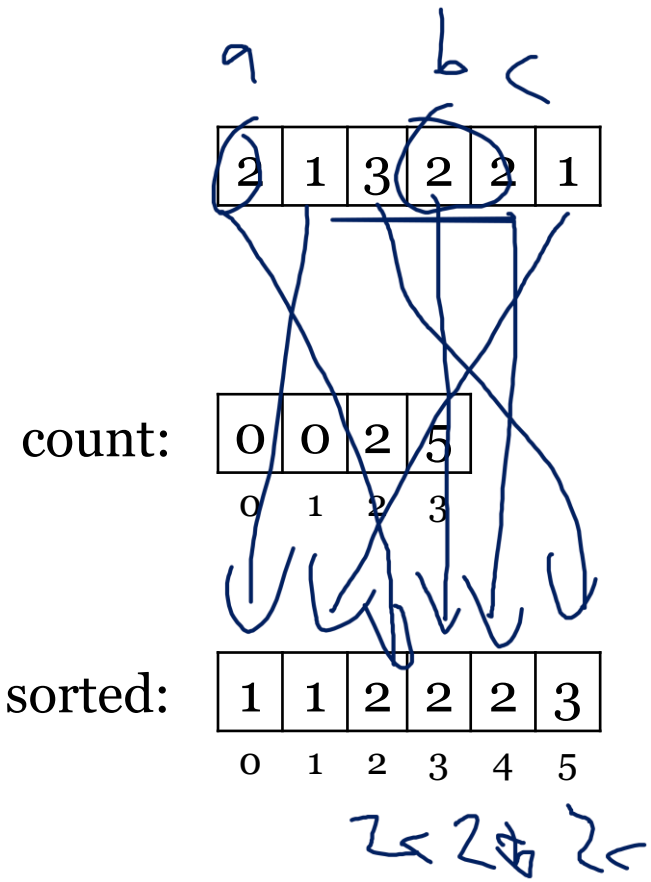
```



CountingSort example

Input: *data*: array of n values
Input: n: size of *data*
Output: a permutation of *data* such that
$data[0] \leq \ldots \leq data[n-1]$

Algorithm: CountingSort
$max = \max(data)$
$count = \text{Array}(max)$
Initialize *count* to 0
for $i = 0$ to $n-1$ **do**
  Increment $count[data[i]]$
end
Perform a cumulative sum on *count*
$sorted = \text{Array}(n)$
for $i = n-1$ down to 0 **do**
  Decrement $count[data[i]]$
  $sorted[count[data[i]]] = data[i]$
end
return *sorted*



CountingSort analysis

Input: *data*: array of n values
Input: n: size of *data*
Output: a permutation of *data* such that
$data[0] \leq \ldots \leq data[n-1]$

Algorithm: CountingSort

$\Theta(n)$ $max = \max(data)$
$\Theta(max)$ $count = \text{Array}(max)$
Initialize *count* to 0
$\Theta(n)$ **for** $i = 0$ to $n-1$ **do**
  Increment $count[data[i]]$
end
$\Theta(n)$ Perform a cumulative sum on *count*
$sorted = \text{Array}(n)$
$\Theta(n)$ **for** $i = n-1$ down to 0 **do**
  Decrement $count[data[i]]$
  $sorted[count[data[i]]] = data[i]$
end
$\Theta(1)$ **return** *sorted*

$\Theta(n + max)$

CountingSort analysis

Worst case complexity

```
Input: data: array of n values
Input: n: size of data
Output: a permutation of data such that
        data[0] ≤ ... ≤ data[n − 1]
Algorithm: CountingSort
max = max(data)                          O(n)
count = Array(max)                       O(max)
Initialize count to 0
for i = 0 to n − 1 do                    O(n)
    Increment count[data[i]]
end
Perform a cumulative sum on count        O(max)
sorted = Array(n)                        O(1)
for i = n − 1 down to 0 do               O(n)
    Decrement count[data[i]]
    sorted[count[data[i]]] = data[i]
end
return sorted                            O(1)
```

Total:
O(n + max)

- Complexity depends on data range
  - Potentially linear
- Good when range is small
- Terrible if range is large
  - Increases memory requirements, too
- Only works with integers or integer-like data

BucketSort

- **Main idea**
  - Divide data range into *n* "buckets"
  - Assign data to buckets and sort independently
  - Buckets "should" take *O(1)* to sort

- **Pseudocode**
  - Calculate max and min of data
  - Create *n* lists
  - For each element *data[i]*,
    - Compute $b = \text{floor}(n(data[i] - \min) / (\max - \min + 1))$
    - Insert *data[i]* into *list[b]* in sorted order
  - Concatenate all lists

BucketSort example

- **Data:**

27	63	30	91	0	13	76	61	99	55

  - 10 elements

BucketSort example

- **Data:**

27	63	30	91	0	13	76	61	99	55

  – 10 elements
- **Data range:** [0, 99]
- **Bucket size:** (max – min + 1) / n = 10
- **Bucket ranges:**

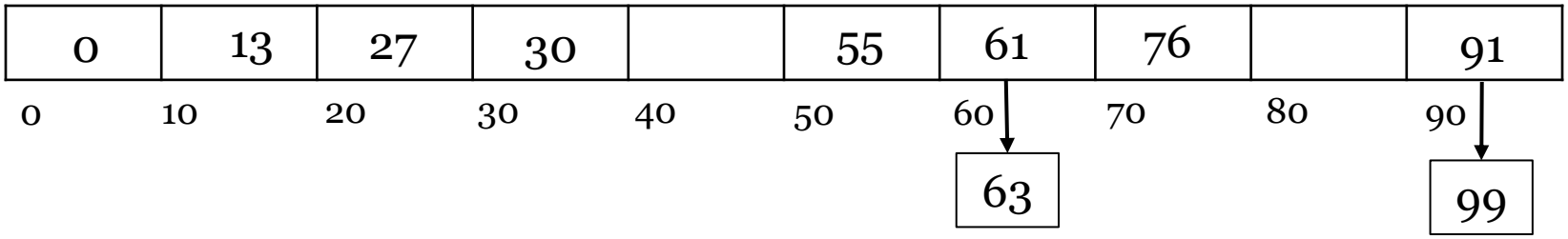
0	13	27	30			63	76		91
0	10	20	30	40	50	60	70	80	90

BucketSort example

- **Data:**

27	63	30	91	0	13	76	61	99	55

  – 10 elements
- **Data range:** [0, 99]
- **Bucket size:** (max – min + 1) / n = 10
- **Bucket ranges:**



- **Solution:**

0	13	27	30	55	61	63	76	91	99

BucketSort analysis

Analysis

- Calculate max and min of data $\Theta(n)$
- Create *n* lists $\Theta(n)$
- For each element *data[i]*, n iters
  - Compute $b = \left\lfloor \frac{n(data[i] - \min)}{\max - \min + 1} \right\rfloor$ $\Theta(1)$
  - Insert *data[i]* into *list[b]* in sorted order $\Theta(list)$
- Concatenate all lists $\Theta(n)$

BucketSort analysis

Analysis

	Complexity
• Calculate max and min of data	$\Theta(n)$
• Create *n* lists	$\Theta(n)$
• For each element *data[i]*,	$\Theta(n)$
– Compute $b = \left\lfloor \frac{n(data[i] - \min)}{\max - \min + 1} \right\rfloor$	$\Theta(1)$
– Insert *data[i]* into *list[b]* in sorted order	$\Omega(1)$ to $O(n)$
• Concatenate all lists	$\Theta(n)$

- Total complexity: $\Omega(n)$ to $O(n^2)$
  - If each list is *O(1)*: $\Theta(n)$
- Linear time if data is evenly distributed
- Stable
- Quadratic time if not
- Uses $O(n)$ space

RadixSort

- **Main idea**
  - Treat numbers as a sequence of digits
    - Also works for words (sequence of characters)
  - Sort digits from least to most significant
    - By the time you reach the last digit, you're done!

- **Pseudocode**
  - For each digit d_i from least to most significant
    - Use CountingSort to sort data by d_i

RadixSort example

- For each digit d_i from least to most significant
  - Use CountingSort to sort data by d_i
- **Example**

data:	27	63	30	91	0	13	76	61	99	55

RadixSort example

- For each digit d_i from least to most significant
  - Use CountingSort to sort data by d_i
- **Example**

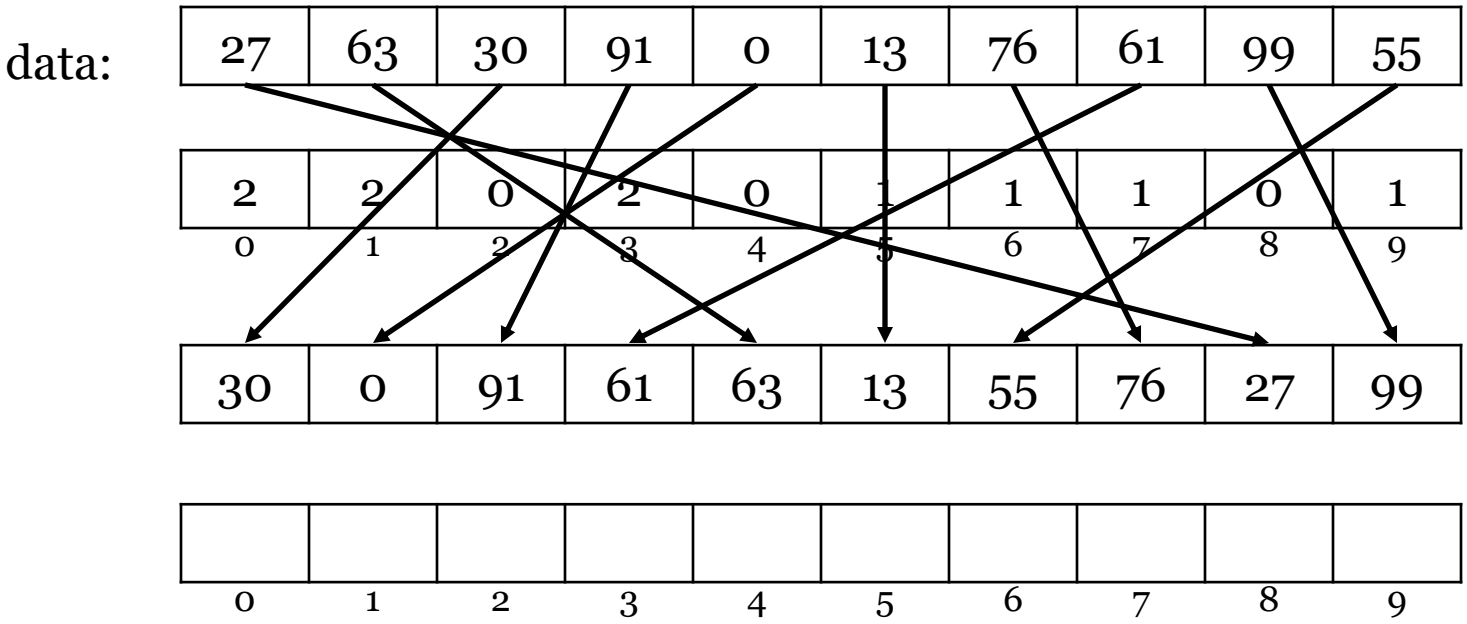
data:

27	63	30	91	0	13	76	61	99	55

0	1	2	3	4	5	6	7	8	9

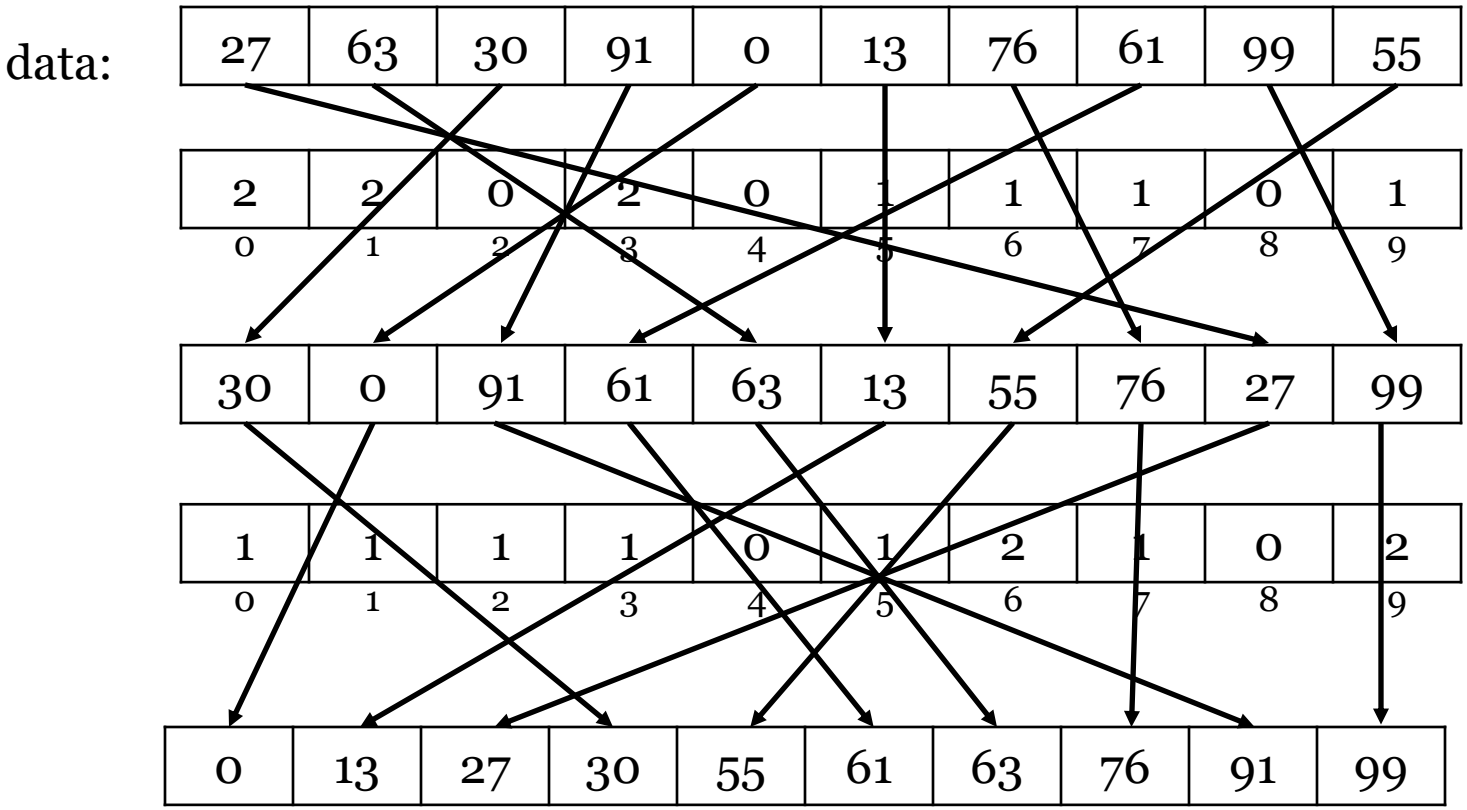
RadixSort example

- For each digit d_i from least to most significant
  - Use CountingSort to sort data by d_i
- **Example**



RadixSort example

- For each digit d_i from least to most significant
  - Use CountingSort to sort data by d_i
- **Example**



RadixSort analysis

Analysis

- For each digit d_i from least to most significant
  - Use CountingSort to sort data by d_i

RadixSort analysis

Analysis

Complexity

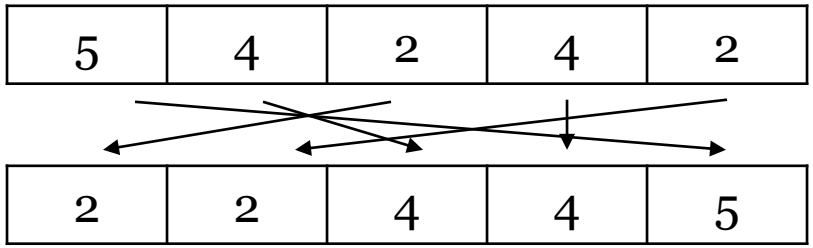
- For each digit d_i from least to most significant — $\Theta(\lg r)$
  - Use CountingSort to sort data by d_i — $\Theta(n)$
- Total complexity: $\Theta(n \lg r)$
  - If $r = O(n)$, complexity is $O(n \lg n)$
  - If $r = O(1)$, complexity is $O(n)$

- Linear complexity if range is bounded
- Uses $O(n)$ space
- Performance depends on range/length of longest entry

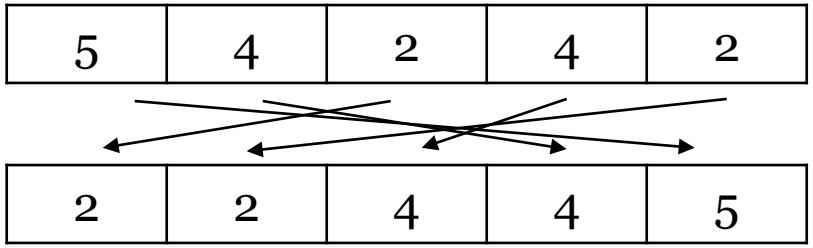
Sorting algorithm evaluation

- There are 3 main features to consider when sorting data
  - **Speed/time complexity**
    - Usually most important factor
  - ***In-place* sorting**
    - Algorithm uses no data structures other than array
      - No temp arrays!
    - Very memory efficient
  - ***Stable* sorting**
    - Equal values retain their relative order
    - *Unstable*: equal values appear in random order

<u>Stable</u>



<u>Unstable</u>



    - Important for some applications
      - E.g., sorting rows in Excel file

Sorting algorithm comparison

	Best-case complexity	Worst case	Avg/exp case	Stable?	In-place?
SelectionSort	$\Omega(n^2)$	$O(n^2)$	$\Theta(n^2)$	✓	✓
InsertionSort	$\Omega(n)$	$O(n^2)$	$\Theta(n^2)$	✓	✓
BubbleSort	$\Omega(n)$	$O(n^2)$	$\Theta(n^2)$	✓	✓
HeapSort	$\Omega(n \lg n)$	$O(n \lg n)$	$\Theta(n \lg n)$	*	✓
MergeSort	$\Omega(n \lg n)$	$O(n \lg n)$	$\Theta(n \lg n)$	✓	✗
QuickSort	$\Omega(n \lg n)$	$O(n^2)$	$\Theta(n \lg n)$	*	✓
CountingSort	$\Omega(n + r)$	$O(n + r)$	$\Theta(n)^\dagger$	✓	✗
BucketSort	$\Omega(n)$	$O(n^2)$	$\Theta(n)^\dagger$	✓	✗
RadixSort	$\Omega(n)$	$O(n \lg r)$	$\Theta(n \lg n)^\dagger$	✓	✗

r: data range
† with "good" data
* possible but slower

Coming up

- Midterm Exam

- **Recommended readings:** Sections 8.1-9.2
- *Practice problems:* R-8.2, R-8.4, R-8.7, C-8.1, C-8.3, C-8.5, C-8.9, A-8.2, R-9.4, R-9.5, C-9.1, C-9.3, C-9.12, A-9.2